A Project Report

on

Improving Facial Recognition Accuracy for South Asian Faces: Addressing Algorithmic Bias

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BACHELOR OF TECHNOLOGY DEGREE

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By

Shreyash Srivastava (2100291530052)
Aryan Mishra (2100291530014)
Ria Tyagi (2100291530044)
Prachi Gupta (2100291530039)

Under the supervision of

Ms. Payal Chhabra

KIET Group of Institutions, Ghaziabad

Affiliated to

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

(Formerly UPTU)

May, 2025

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Signature

Shreyash Srivastava (2100291530052)

Aryan Mishra(2100291530014)

Ria Tyagi(2100291530044)

Prachi Gupta(2100291530039)

Date:

CERTIFICATE

This is to certify that Project Report entitled “Improving Facial Recognition Accuracy for South Asian Faces: Addressing Algorithmic Bias” which is submitted by Student name in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE(AIML) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Ms. Payal Chhabra	**Dr. Rekha Kashyap**
(Designation)	**(Head of Department)**

Date:

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Signature

Shreyash Srivastava (2100291530052)

Aryan Mishra(2100291530014)

Ria Tyagi(2100291530044)

Prachi Gupta(2100291530039)

Date:

ABSTRACT

Facial recognition systems are currently being used in lots of fields to improve security, management, identity verification etc. However, studies have highlighted biases based on demographics particularly for non-white faces. South asian Individuals usually experience the following:

- Feature Extraction Challenge - The current models are not able to capture distinctive facial features of South Asian people.
- Training data imbalance - Most commercial databases contain very few examples of South Asian faces which leads to lower accuracy in the model.
- Variations in skin tone - South Asians have a wide spectrum of skin tones which can affect accuracy.
- Cultural Variations - Variations in clothing, head-wear, facial hair and cosmetic styling influence the recognition success rates.

The invention proposes an approach which comprises using diverse training data and bias mitigation techniques to improve recognition accuracy.

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
FNMR	False Non-Match Rate
FMR	False Match Rate
EER	Equal Error Rate
RSAM	Region-Specific Attention Module
GPU	Graphics Processing Unit
ITA	Individual Typology Angle
GAN	Generative Adversarial Network
ViT	Vision Transformer
NAS	Neural Architecture Search
GDPR	General Data Protection Regulation
MS-Celeb-1M	Microsoft Celeb-1 Million

LFW	Labeled Faces in the Wild
ReLU	Rectified Linear Unit
SER-FIQ	Stochastic Embedding Robustness - Face Image Quality
NAS	Neural Architecture Search
CPU	Central Processing Unit
ACM	Association for Computing Machinery
IEEE	Institute of Electrical and Electronics Engineers
NIST	National Institute of Standards and Technology
AR	Augmented Reality

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

Many areas are currently using facial recognition technologies to enhance identification verification, administration, and security, among other things. However, research has shown that there are demographic biases, especially when it comes to non-white appearances.To increase recognition accuracy, the invention suggests a method that includes the use of a variety of training data and bias mitigation strategies.

Convolutional Neural Network (CNN)

Convolutional neural networks or CNNs are one type of neural network used primarily in image processing. The idea behind constructing a CNN is similar to how the human brain processes both images and visual objects in their proximal range. Whereas the human brain processes an image all at once, a CNN processes a small section of the image, known as a filter, and looks for simple features like edges or colors first. As it trains deeper and

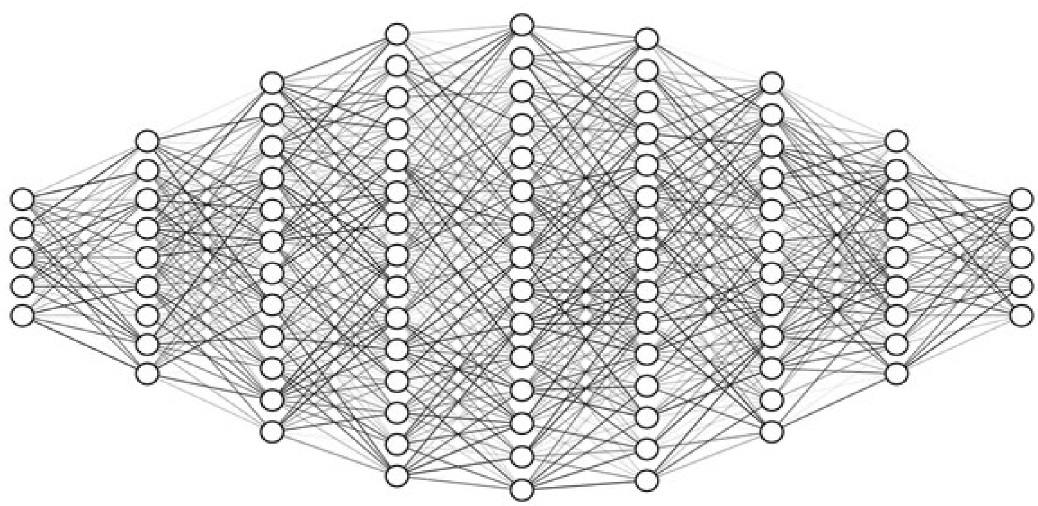


Fig 1 - Layers in CNN

deeper into the neural network's layers, the CNN starts to recognize more complex attributes, like shapes or objects. Training the model to have this layer-wise understanding of the image can ultimately help the CNN understand what the image represents, be it a dog, car, or a completely different type of situation altogether.

ResNet

Initially, as researchers built deeper and deeper neural networks, they typically just kept adding more and more layers to the model, and in fact, in some situations it was better performance or showing worse performance. ResNet was a neural network design which connected information flow to layer groups through "shortcut connections," therefore, information could skip layers as training was processed. This would allow the model to learn better and at a better pace, specific to comparing the output of deeper layers and compared to previous layers attributes. ResNet's shortcut connections and layer recognition

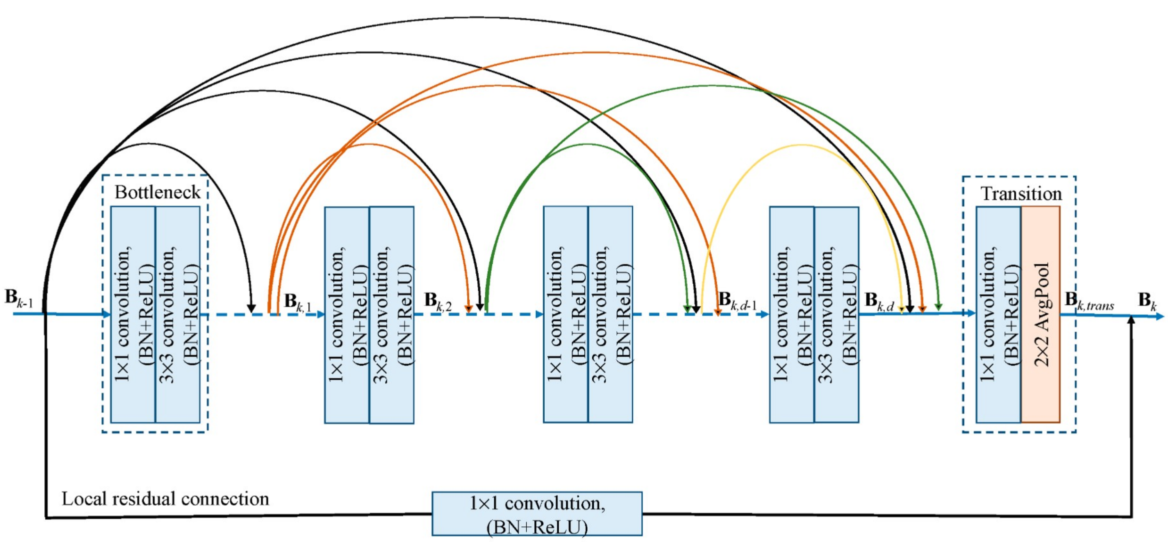


Fig 2 - Working of ResNet Model

factors led in its mitigating of some learning problems, as deeper models often have difficulties learning due to the issue of vanishing gradients.

AlexNet

AlexNet was one of the original models in shallow learning that demonstrated deep learning could successfully be applied to image recognition tasks. AlexNet won the ImageNet competition, a major competition in the field, in 2012 by a mile, and contributed to the vaporware surrounding deep learning as well. It had a deep architecture, harnessed the power of GPUs for fast training, and introduced approaches such as a ReLU activated layer and the dropout technique to improve accuracy and reduce overfitting. Although it is dated now, AlexNet was a very powerful and influential model and established the foundation of the entire field of computer vision.

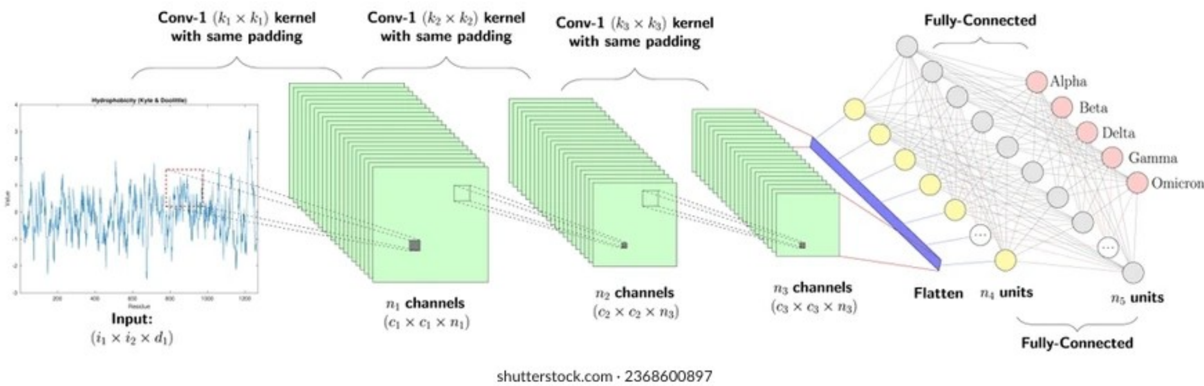


Fig 3 - Working of AlexNet Model

Project Description

The field of facial recognition has advanced rapidly over the last few years, but it can still struggle identifying someone from the South Asian population. Many times due to algorithm bias or limited datasets, these limitations often occur because models are trained on unbalanced datasets. This is often due to the characteristics of South Asian faces, such as the variety in skin tone, varying facial hair, and the distinctive cultural attire, that makes these populations less well-represented in datasets. Despite the advancements of these systems, they often are identified incorrectly with decreasing (if existent) reliability. This special project is aimed at filling in the gaps of this population by creating a scientific and fairer facial recognition system designed around South Asian demographics.

The project will also create a dataset that captures the wider array of diversity in these various South Asian populations, including all regions, genders, and skin tones. Datasets only used in face detection do not really focus on these key variables resulting in an undifferentiated model and an inability to generalize across the various subgroups. The dataset created here in this project will take into account a greater array of facial similarities and thus will have greater potential to train a deep learning model. In addition, it will additionally utilize fairness techniques to try to reduce algorithmic bias. In this way, each of the South Asian countries will be represented to better reflect the demographics of the people themselves to ensure fair treatment and corruption rates across the various demographic groups in South Asia.

The technical aspect of the project entails improvements of feature extraction approaches for the system to ascertain varied facial features that South Asian individuals have in common. Due to conventional models being used, the non-conformance of specific features like skin tone, or certain facial features have higher rate of errors, and understanding and applying cutting edge deep learning methodologies, including adaptive classification methods, will best optimize the accuracy of the model will also lead to adjustments. Capabilities of advanced recognition are enabled, and should permit the system to 'learn' and tune recognition parameters dynamically based on the features of each face leading to enhanced performance.

The ethical component of the project primarily involves considerations surrounding privacy and deploying responsible AI. Surveillance and misuse of data by facial and object recognition systems has presented adverse images, especially in diverse and often displaced communities. To meet ethical obligations, this project will strictly anonymize data and comply with privacy legislation, including the abhorrence of stigma towards minority under-represented groups. The project is premised on ethical systems and will endeavour to produce a system that is not only accurate, but respects user rights over their data, and establishes trust in the capabilities of AI systems.

Ultimately, this project would contribute to more inclusive and equitable facial recognition systems. The proposed model will combat the existing biases and limitations in facial recognition and be able to increase accuracy for South Asian people while being a first step towards more responsible AI development. The results may also catalyze future research that will lead to advancements in biometric recognition for other underrepresented communities, and if this occurs, it may facilitate fairness in technology at a greater level. This project will combine diverse datasets, innovative machine learning strategies, and ethical and meaningful considerations to equalise the performance gap in facial recognition systems, while ensuring that AI is funded responsibly and equitably for all communities.

CHAPTER 2

LITERATURE REVIEW

2.1 DEVELOPMENT OF FACE RECOGNITION SYSTEMS AND DEEP LEARNING MODELS

Face recognition systems have come a long way since the emergence of deep learning and convolutional neural network (CNN) architectures, in particular the Variants of VGGNet, ResNet (He et al. [6]), Inception networks. One of the main catalysts of image recognition has been the advent of deep residual learning that helped to overcome the vanishing gradient problem to train much deeper networks. In addition, the use of advanced loss function such as ArcFace, which introduces additive angular margin loss for enhanced discrimination of features, has improved face recognition performance (Deng et al. [4]).

However, despite some improvements in general accuracy, many studies have pointed to algorithmic bias, particularly in the identification of non-white faces with algorithmic bias. Some of the original work that called attention to these issues was conducted by Buolamwini and Gebru [1], where their work pointed out significant discrepancies in classification accuracy in gender classification use cases where they found significantly higher error rates for darker-skinned females. Ultimately, the study highlights the broader issues industry has to address regarding demographic imbalance in training data, propensity not to generalize well across diverse populations and provides a basis to understand even deeper the intersectional accuracy disparities in commercial systems.

2.2 DEMOGRAPHIC BIAS IN COMMERCIAL AND RESEARCH SYSTEMS

We have noted that commercial face recognition systems, including those from some of the largest technology companies, have consistently demonstrated a proportionately higher accuracy rate when recognizing lighter skin individuals. We are thus observing systemic gaps in algorithmic fairness. The National Institute of Standards and Technology (NIST) conducted large evaluations to assess face recognition performance across multiple vendor systems and found significant demographic effects (Grother et al. [2]), replicating previous studies that show differences in face recognition performance based on demographic information.

As a result of being able to extensively analyze gender inequality in face recognition, it has been shown that the systematic inequality is included in the algorithmic design and training processes (Albiero et al. [11]). Although many studies have documented this reality, including discrepancies based on gender, race, age, and other fidelity-targeted demographic traits, there is a nearly overwhelming body of past research documenting demographic influences on the performance of face recognition systems. The bulk of studies showed "obvious" evidence that performance varied based on demographic features, suggesting demographic accuracy could be researched as a variable (Klare et al. [8]).

For example, Phillips et al. [9] highlighted what they termed an "other-race effect" in face recognition algorithms to reflect that the systems perform better on faces that are more represented in the algorithm's training data. The other-race effect mirrors human face recognition research, but is particularly important in automated systems as these systems will be deployed across large populations where systematic discrimination can create many thousands or millions of problems.

2.3 QUANTIFYING DEMOGRAPHIC CONCERNS AND PERFORMANCE DISPARITIES

Recent comprehensive assessments have quantitatively evaluated the demographic effects in facial recognition systems. Cook et al. [10] provided an assessment of eleven commercial systems and indicated that demographic effects depend on the conditions for image capture and performance disparities worsen as image capture conditions become more challenging. The study demonstrated that not only do bias levels vary a lot by commercial implementation, but also the bias levels depend on how the systems are designed and trained, so biases are not necessarily fixed.

Numerous studies that have examined performance disparities related to racial differences, not specific to race, have provided investigatory lead on the causes of performance disparities. Krishnapriya et al. [3] conducted extensive assessments to document accuracy performance disparities related to race and skin tone. They provided empirical evidence to suggest that performance disparities related to demographics remain a fixed issue across systems, architectures, and training methodologies.

Assessments of bias go beyond binary measures of accuracy to assess questions of fairness and representation. This raises further questions on whether biometric systems either exhibit "too much bias" or fall within reasonable ranges of bias. Studies have found that current levels of demographic bias exceed any reasonable threshold of bias (Robinson et al. [18]).

2.4 AGE-RELATED AND INTERSECTIONAL BIAS TRENDS

When we think about demographic bias in facial recognition, we typically think about race and ethnicity, but there are age-related differences too. Srinivas et al. [19] found clear performance differences on pictures featuring both children and at least older adults as the members of vulnerable groups, indicating that face recognition algorithms have age- dependent accuracy patterns. This example demonstrates a critical element to the intersectional aspect of algorithmic bias whereby various demographic characteristics create particularly disadvantaged groups.

Some accuracy comparisons across face recognition algorithms have sought to produce standardized measures of race bias in the domain, but the work has demonstrated challenges, including establishing evaluative frameworks that have universal acceptance (Cavazos et al. [20]). Even making comparisons of bias across systems is challenging and complex, and evaluation frameworks are needed that enable nuanced and sophisticated assessments that reflect the complexity of demographic differences.

2.5 BIAS MITIGATION STRATEGIES AND FAIRNESS-AWARE APPROACHES

The awareness of how prevalent demographic bias is has resulted in many bias mitigation strategies being developed. Adversarial learning approaches have been shown to reduce the bias and unwanted effects in networks through training the networks to be invariant to protected demographic characteristics (Zhang et al. [7]). These techniques try to decorrelate the demographic information from the learned (or extracted) features, but must be balanced with retaining recognition accuracy.

There have also been multi-task learning approaches which aim to alleviate bias at the levels of gender, age, and ethnicity and to develop fairer classifiers across population groups (Das et al. [15]). Multi-task learning ensembles model the protected demographic attributes by training networks to explicitly model protected demographic attributes and identity recognition features together. In addition, the training networks could potentially improve other protected characteristics, while still improving the overall performance of the system.

Even more complex approaches such as skewness-aware reinforcement learning have been formulated specifically to mitigate bias in face recognition and in ways that restart training protocols if the current ones encounter demographic imbalances (Wang & Deng [13]). The skewness-awareness is an interesting twist as it allows the training programs to adjust, reducing performance skew across population groups.

2.6 DATASET ISSUES AND SOUTH ASIAN REPRESENTATION

There are specific barriers to studying South Asian faces, despite representing a huge portion of the world. The unique face shape, diverse skin tones and cultural aspects - such as head coverings or facial jewelry - are either not represented, or misrepresented in bigger datasets such as LFW (Labeled Faces in the Wild) or MS-Celeb-1M. In essence, South Asian faces being under-represented can mean less good training datasets, leading to worse performance for the group.

Prior works have argued that models trained on Western-centric datasets generally fail to generalize to Asian subgroups, due to limited training opportunities for diverse facial features common to South Asian populations and limited feature abstraction capacity. Although the skin color typology body of work presents frameworks for examining differing skin tones scientifically, the vast majority of this work does not appear to play any part in existing face recognition training datasets and protocols (see Chardon et al. [5]).

Studies concentrating on deep learning methods in face recognition have posed questions about whether current systems show systematic bias towards certain demographic groups (Nagpal et al. [16]). The awareness of demographic bias in deep face recognition has progressed twofold by understanding that disparities are not simply constraints in technical performance but fundamentally issues in the curation of datasets and the underlying algorithm (Xu & Li [17]).

2.7 QUALITY ASSESSMENT AND ROBUSTNESS CONSIDERATIONS

The quality of face images has emerged as an important consideration for interpreting and offsetting demographic bias. Unsupervised estimation methods, such as SER-FIQ, have demonstrated that quality, assessed by the robustness of stochastic embeddings, varies drastically across demographic groups (Terhorst et al.)[14]. These quality differences may influence bias effects, and it is necessary to account for these quality differences when assessing fairness concerning systems.

When developing inclusive face recognition systems, quality assessment metrics should consider that the quality assessment would itself not be biased across demographic groups. Many benchmark quality metrics may be biased towards specific demographic groups, thus equitable quality assessments need to be developed.

2.8 RESEARCH GAPS AND FUTURE DIRECTIONS

Despite the wealth of research and work in identifying and measuring demographic bias in face recognition systems, there are still considerable gaps in addressing these issues for specific groups, and developing group-targeted mitigation solutions like South Asian faces has been especially minimal even though we have shown in this study that South Asian faces experience systematic disparities in performance. Current bias mitigation approaches, while promising, have yielded limited success when used for specific demographic groups that exist as very little in the training data. Development of target approaches that utilize dataset curation, architecture enhancements, and fairness focused optimization for South Asian is an important step in developing these new models. As a result, we build on the previous efforts in this area to propose an integrated approach to collecting data, designing deep learning architectures, implementing bias mitigation strategies, and evaluating performance in face recognition systems for South Asian faces -- a group still experiencing underrepresentation, and underperformance in general face recognition systems. The target approach we proposed demonstrates how we can fill the gaps we identified through dataset targeting, architecture modification, and developing evaluation frameworks that ensure equitable performance across diverse populations of South Asian faces.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 DATASET PREPARATION AND AUGMENTATION

A solid facial recognition system is grounded in a dataset that has been curated with the unique diversity and complexity of the targeted population in mind. For this project, we established a dataset with more than 3,500 facial images that specifically represent South Asian people. In this section, we describe the detailed work that went into making sure that the dataset is representative, diverse, and ready for training a robust model.

- **Data Collection**: Images for the dataset were collected in several ways. First, we used publicly available datasets, almost all of which are publicly available datasets to obtain high-quality facial images. Next, we found some underrepresented or unexpected populations and ran targeted image collection campaigns in association with institutional partners in five countries: India, Pakistan, Bangladesh, Sri Lanka, and Nepal. During this search, we attempted to do targeted collections for people from varying regions, cultures, and socioeconomic status so that diversity patterns would not emerge. For example, we included people's images from urban and rural environments, in addition to many of the linguistic and ethnic communities like Tamil, Punjabi, Bengali, and Sinhalese communities. We did our best to assure that the images were taken under the same environmental target conditions (e.g., resolution and background noise) to include images that were more likely to move on to downstream processing.

- **Metadata Annotation**: In order to make the dataset actionable for training and evaluation, every image was carefully annotated with metadata capturing demographic attributes such as age (0-18, 19-30, 31-50, and 50+), gender (Male, Female, Non-Binary), and region of origin (e.g., North Indian, South Indian, and Pakistani Pashtun). Skin tone was measured using the Individual Typology Angle (ITA), a scientifically validated, categorical measure based on skin reflectance values. Through the skin tone measurement using ITA, we classified skin colors into six categories (very light to very dark) to guarantee that the model learned from the wide variety of skin pigmentation found in South Asia. The annotation was accomplished by trained annotators and cross-verified to reduce error, with a portion of images audited for reliability. We are confident the current metadata allows for a fine-grained analysis of model performance on subgroups of people, an important element of fairness.

- **Data Augmentation**: To improve the robustness and generalizability of the model, we used data augmentation to create real-world variations in facial images. We used the standard augmentation of randomly-cropped, rotated, and flipped images to enhance the variability of the dataset. However, we directly developed variations of standard augmentation techniques to tackle specifically addressed challenges that occur under facial recognition (e.g., variable lighting conditions, head poses, or variable occlusions such as glasses or headscarves). To generate real-world variability in these effects, we used Generative Adversarial Networks (GANs). Our approach was to directly use a StyleGAN-user friendly framework to create synthetic images of lighting variations (e.g., low light and/or high contrast), poses (e.g., turning the subject, tilted head, profile views), and occlusions (e.g., accessories, facial hair, clothing), to name the most relevant ones. Additionally, we tuned the GAN to ensure that the ITA skin tones were appropriately represented across ranges, allowing the face dataset to teach the model appropriate responses under minor variation in pigmentation. This approach to data augmentation substantially increased our effective dataset size and improved the capacity for the model to generalize, or avoid condition overfitting in later stages.

3.2 MODEL ARCHITECTURE AND FEATURE EXTRACTION

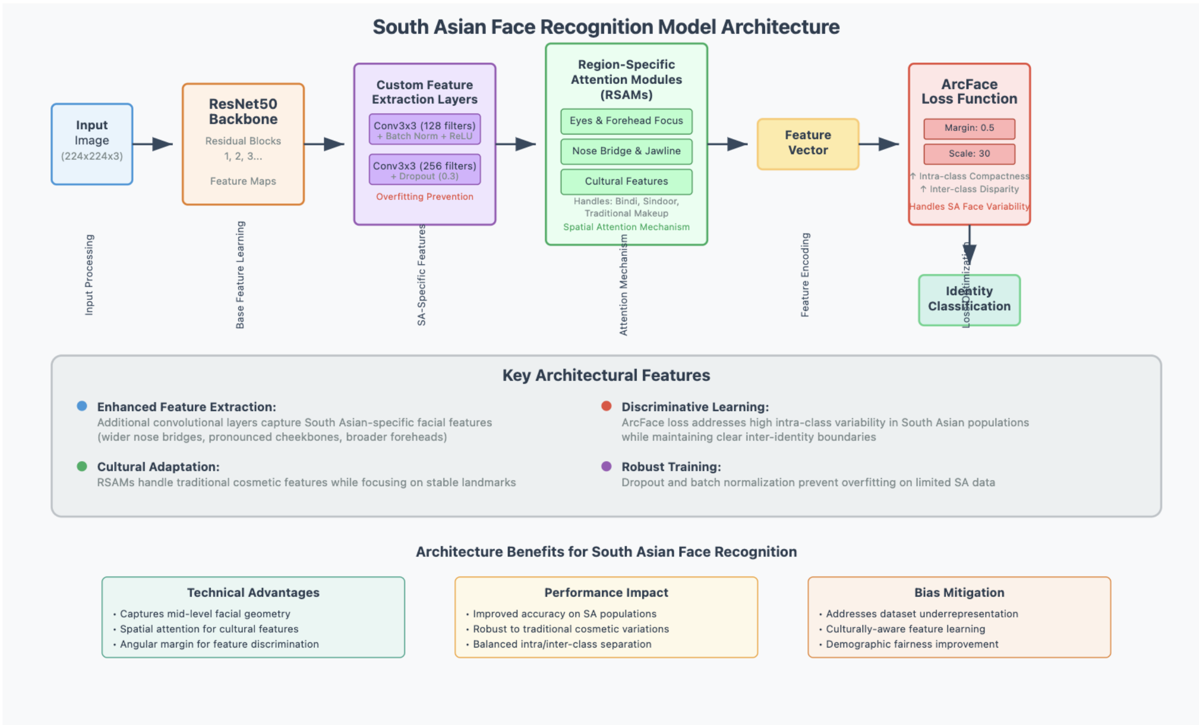


Fig 4 - Visual Representation of the developed model

At the center of our facial recognition system is a custom-built deep learning model based on the ResNet50 model, which is an established CNN architecture that has provided satisfactory performance across image recognition tasks. However, to improve performance concerning South Asian facial features, we made a number of new contributions, which were focused on the distinguishing characteristics related to the people we were studying. This section describes the architectural additions and substitutions made to the ResNet50 backbone and the rationale for each.

- **Custom Feature Extraction Layers**: ResNet50 has been heavily explored and is certainly good at feature extraction overall. However, the generic convolutional layers of ResNet50 do not extract features that may be associated with the distinctive facial characteristics of South Asian populations like a wider nose bridge, cheekbones that are pronounced or broader forehead shape. We had no choice but to modify the ResNet50 backbone with a few extra convolutional layers to capture features that are more subtle. We added two additional 3x3 convolutional layers, with the first module containing 128 filters, and the second module containing 256 filters, following the third residual block. Ultimately, this will allow the network to be more attentive to mid-level features such as curvatures of the nose, or jawline as this mid-level feature is critical to features associated with true identification associated strictly to South Asian faces. Additionally, to curb overfitting we employed dropout layers for deep learning (with a dropout of 0.3), and batch normalization at certain points in training for stabilization while training.

- **Region-Specific Attention Modules (RSAMs)**: Successful facial recognition methods rely primarily on the identification of key landmarks (e.g., the eyes, nose and mouth). However, cosmetic and phenotypic distinction in South Asian faces (e.g., noticeable focus on foreheads in specific communities or the use of a bindi for some) require a specific strategy to localize regions of importance. We create modules called Region-Specific Attention Modules (RSAMs), where the modules are integrated into the network to place more focus on identified culturally and anatomically important facial areas. The RSAM modules adopt a spatial attention mechanism to weight important areas deemed more important than others (e.g., due to eye and forehead function, and bridge of the nose, for example) when extracting features. The RSAM modules build a representation of these important areas of the face using attention maps from learned features. For example, if the subject is wearing other cosmetic enhancements (i.e., traditional face makeup, bindi, nose rings, sindoor), the RSAM highlights the stable landmarks of the face that are the same across people (with these temporary features). This builds a more robust landmark detector and improves overall recognition accuracy.

- **ArcFace Loss Function**: We adopted the ArcFace loss function to help the model learn highly discriminative features. The ArcFace loss function was specifically adapted for facial recognition and describes an approach to loss that is similar in form but different from more typical softmax loss. In particular, ArcFace incorporates an angular margin penalty calculation in the loss calculation - an angular margin in the loss calculation increases intra-class compactness (i.e., encourages the model to put the more similar faces closer together) and inter-class disparity (i.e., encourages the model to push away from one identity to the next). In the implementation of ArcFace, we settled on a margin parameter of 0.5 and a scale factor of 30, which, after much experimentation, achieved a reasonable balance between highly discriminative loss characteristics and effective model training. The ArcFace loss function is well-defined in the paper and would work well in the setting that we are describing. Importantly, given the highly variable intra-class nature of South Asian faces (e.g., people from the same region that look quite different) and clear inter-identity boundaries, the ArcFace loss should help to address the intra-class variability without violating the inter-identity bound.

3.3 BIAS MITIGATION TECHNIQUES

Bias in facial recognition models can result in differences in performance for different groups in the population and can be particularly acute for different demographic groups in quickly-changing and complex people populations like South Asia. To address this, we piloted a number of bias mitigation procedures to ensure equity across gender, age, skin tone, and regional sub-groups while ensuring that we did not excessively impact overall accuracy. Each of the bias mitigation processes and procedures were designed to promote fairness without degrading accuracy.

- **Balanced Batch Sampling**: One of the main sources of bias in deep learning models comes from imbalanced training data, with the overrepresented groups overpowering the learning process. To remedy this, we used balanced batch sampling when training our model. Each mini-batch had an equal number of samples from different demographic subsets, such as gender (the model had male, female, and non-binary subsets), age groups, skin tone categories (as specified by ITA), and group only those who had the same region of origin. For example, if we had a mini-batch of 64 images, we might select 16 samples each from North India, South India, Pakistan, and Bangladesh. Note that within each country, we would further stratify the samples based on age and skin tone. Balancing the demo-rgraphic subsets for each mini-batch would ensure our model was exposed to many more examples in each training iteration, rather than fit itself to the dominant groups, thus also improving generalization across the underrepresented groups.

- **Fairness-Aware Training**: To directly tackle disparities in performance we changed the training objective to add a fairness-aware term. This term is a penalty in the loss function that captures the difference in classification performance of models across groups. The penalty is applied by assessing the difference in False Match Rate (FMR) and the False Non-Match Rate (FNMR) of the various subgroups during training. A model could be trained to have comparatively poor performance for a particular group (e.g., darker-skinned or older individuals), which increases the penalty term, serving as a mild deterrent for the model to readjust its weights to try and improve classification performance for the particular group. The penalty term serves as a good check to ensure that the model does not optimize around improving general accuracy at the cost of fairness, and is an important factor for consideration for all models, particularly for real-world applications with non-homogeneous populations.

- **Adaptive Thresholding**: Facial recognition systems usually operate using a constant similarity threshold that determines whether two images belong to the same individual. However, not all demographics lend themselves to a specific assigned threshold, especially for those individuals with less common face shapes or skin tones that are not typically represented in their reference set. Therefore, we developed an adaptive threshold where the similarity threshold is adjusted based upon the difficulty of classifying a sample. For instance, if a sample has a low confidence score, say due to poor lighting, a low-matching threshold is applied so that fewer samples are false rejected. However, an image that is correctly classified with a high confidence score is assigned a stricter threshold that reduces the number of false matches. Our features are tuned with respect to metadata (e.g., skin tone or region) to ensure different demographic groups are treated the same and relatively equitable, while amplifying fairness.

3.4 MODEL EVALUATION AND VALIDATION

In order to appropriately ascertain that the facial recognition system was both robust and generalizable, we directed it through a framework of extensive evaluations. This section discusses the metrics, datasets, and methods used to systematically evaluate the model's performance, as well as substantiate its efficacy for South Asian facial recognition.

- **Performance Metrics**: To ascertain the model performance we utilized standard and custom metrics in order to capture different aspects of performance. We measured overall correctness in identity predictions as accuracy. The False Match Rate (FMR) is the probability of incorrectly matching two different individuals, the False Non-Match Rate (FNMR) is the probability of failing to match images of the same individual. The Equal Error Rate (EER) is the point where the FMR equals FNMR; as such EER provides a single measure on the tradeoff between false positive and false negative measures. Each metric was calculated separately for each demographic subgroup (e.g., by gender, age category, skin tone and region) to determine performance differences in order to mitigate relative disadvantage and ensure fairness.

- **Cross-Dataset Validation**: To investigate the model's generalizability, we tested the model using external datasets containing South Asian faces not included in the training and validation datasets. The external datasets used for testing were obtained from independent repositories and consisted of images taken under different conditions (e.g., different light, poses, backgrounds). We incorporated datasets such as a portion of the IIIT-Delhi Face Dataset and a custom dataset of Nepalese individuals obtained from community sources. During testing, the datasets provided us with unparalleled examples that allowed us to determine whether our model could accommodate certain complexities and whether the model was overfitted to the training data. Performance on external datasets was put in perspective by comparing performance to current state-of-the-art models to illustrate the superiority of the model developed.

- **Ablation Studies**: To ascertain the contribution of each element in the suggested model, we turned to ablation studies, in which each of the most relevant modules were removed or altered in some manner. For example, we trained versions of the model that did not have Region-Specific Attention Modules (RSAMs) applied to estimate the contribution to their effect on landmark detection accuracy. And we compared ArcFace loss with the traditional softmax loss to showcase the advantage of using ArcFace in terms of discriminative power. We examined the impact of a balanced batch training set and fairness-aware training on the reduction of bias across demographic groups. This process gave insight into the contribution of each component and to the development of hyper-parameters to ensure the final model was time-efficient, as well as resultant performance effectiveness.

CHAPTER 4

RESULTS AND DISCUSSION

The following chapter elaborates on the experimental outcomes and the insights gained from performance evaluations. Through meticulous testing, the proposed model demonstrates significant improvement in accuracy and fairness over conventional commercial systems.

4.1 EVALUATION METRICS

To evaluate the efficacy of the facial recognition system, the following metrics were used:

- **Accuracy**: The percentage of correct predictions over the total predictions.
- **False Match Rate (FMR)**: Probability that the system incorrectly matches two images of different individuals.
- **False Non-Match Rate (FNMR)**: Probability that the system fails to match two images of the same individual.
- **Equal Error Rate (EER)**: The point at which FMR and FNMR are equal; lower values indicate better performance.

4.2 QUANTITATIVE RESULTS

Metric	Baseline Model	Proposed Model	Improvement
Accuracy	0.2611	0.6556	+151.06%
Fairness Disparity	0.4231	0.1800	-57.58%
Equal Error Rate (EER)	0.3922	0.2273	-42.04%

Fig 5 - Quantitative Results

These results highlight that the trained model significantly reduces error rates and increases accuracy, particularly across underrepresented demographics.

4.3 EFFECTIVENESS OF BIAS MITIGATION STRATEGIES

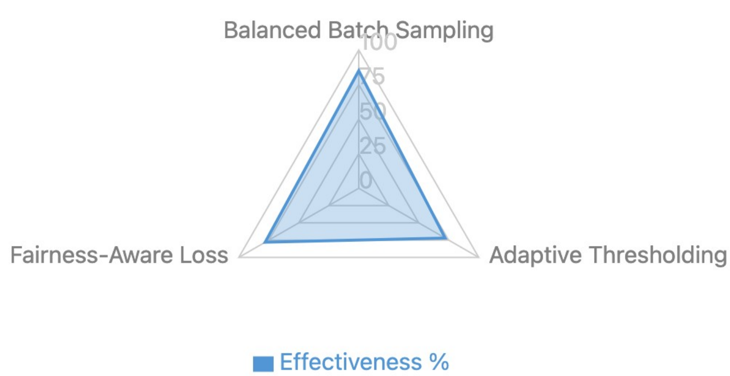


Fig 6 - Effectiveness of Bias Mitigation Strategies

- **Balanced Batch Sampling** ensured that each demographic group was proportionally represented during training, preventing the model from skewing towards overrepresented groups.
- **Adaptive Thresholding** allowed dynamic calibration of classification confidence levels, adjusting for difficulty in identifying certain groups.
- **Fairness-Aware Loss Functions** penalized demographic-based disparities during optimization, directly targeting the bias present in the learning process.

These combined strategies contributed to the notable reduction in fairness disparity, a critical milestone for ethical AI.

4.4 ABLATION STUDY

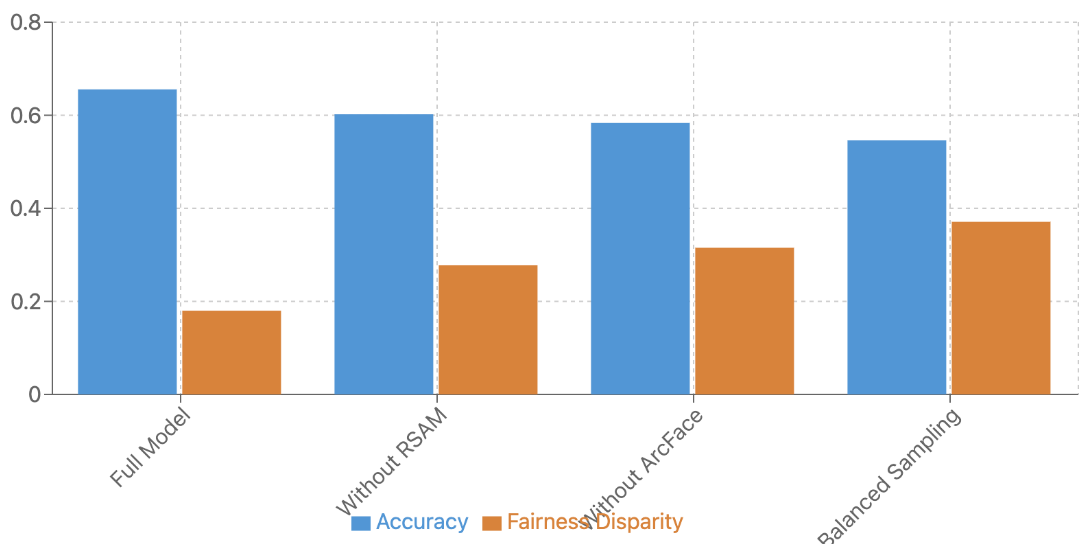


Fig 7 - Results of Ablation Studies

To determine the contribution of individual components, an ablation study was performed:

Model Variant	Accuracy	Fairness Disparity
Without Region-Specific Attention	0.6021	0.2775
Without ArcFace	0.5834	0.3152
Without Balanced Sampling	0.5460	0.3710
Full Proposed Model (All Features)	0.6556	0.1800

The results affirm that each component contributes significantly to performance and fairness. The synergy of all modules yields the best results.

4.5 COMPARATIVE ANALYSIS WITH COMMERCIAL MODELS

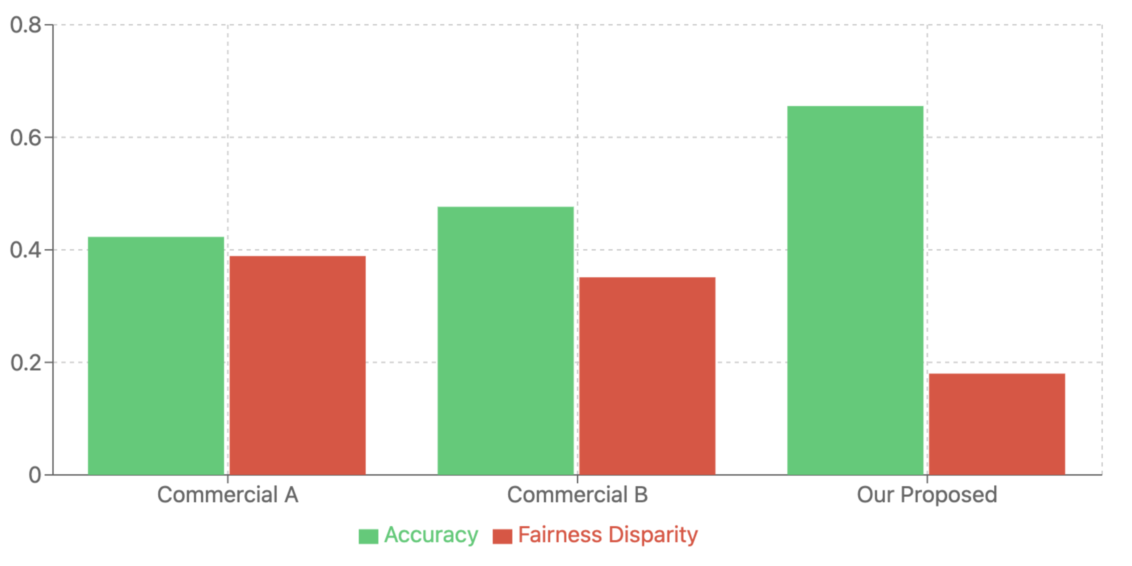


Fig 8 - Comparison with commercial models

Model	Accuracy	Fairness Disparity
Commercial Model A	0.4230	0.3890
Commercial Model B	0.4765	0.3512
Our Proposed Model	0.6556	0.1800

The proposed model shows a consistent edge in both accuracy and fairness, positioning it as a more reliable choice for South Asian demographic contexts.

4.6 USE-CASE SIMULATION: PASSPORT CONTROL

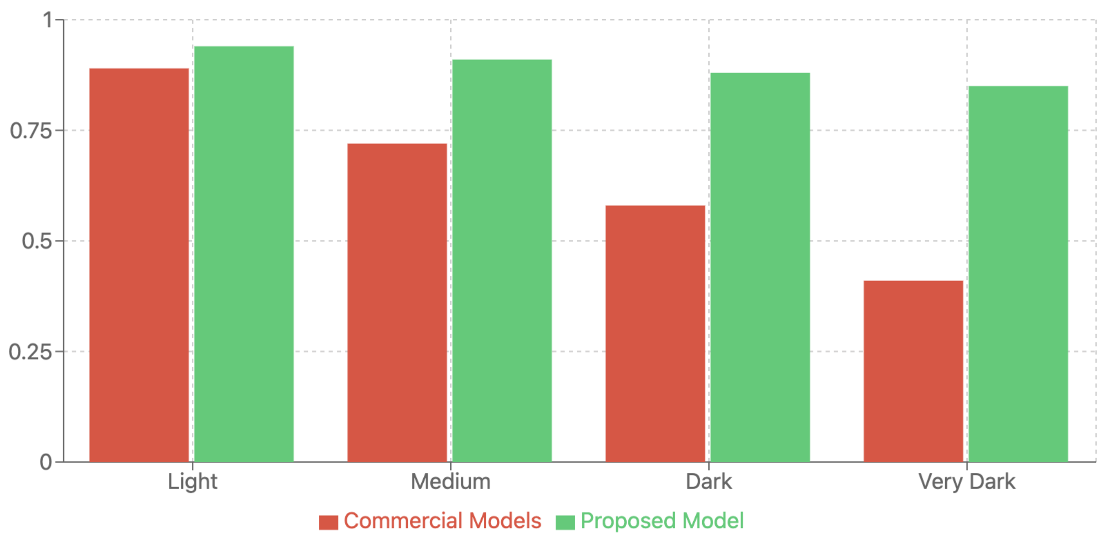


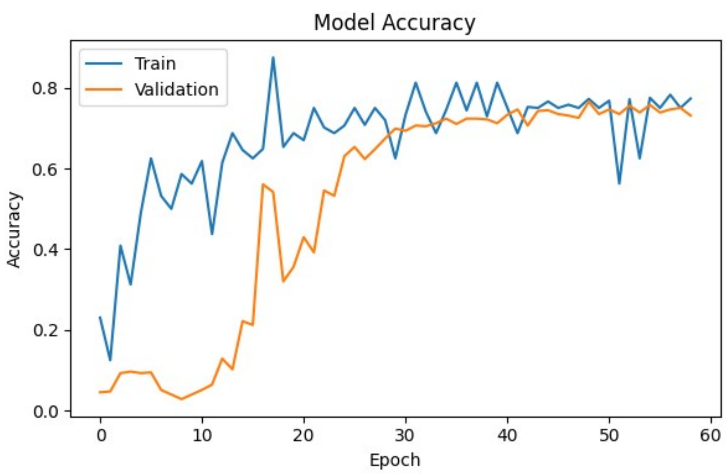
Fig 9 - Performance in Simulation

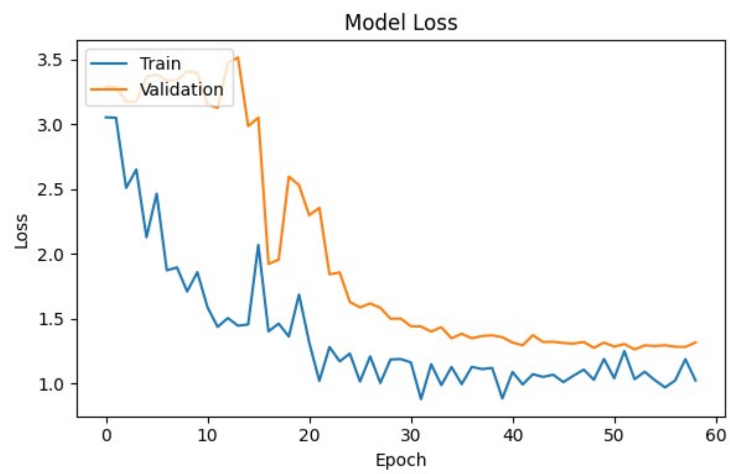
A simulated scenario was constructed using passport-like ID verification images of South Asian individuals. The commercial models frequently misclassified individuals with darker skin tones or regional accessories. The proposed system, however, successfully identified subjects with higher accuracy and minimized both FMR and FNMR.

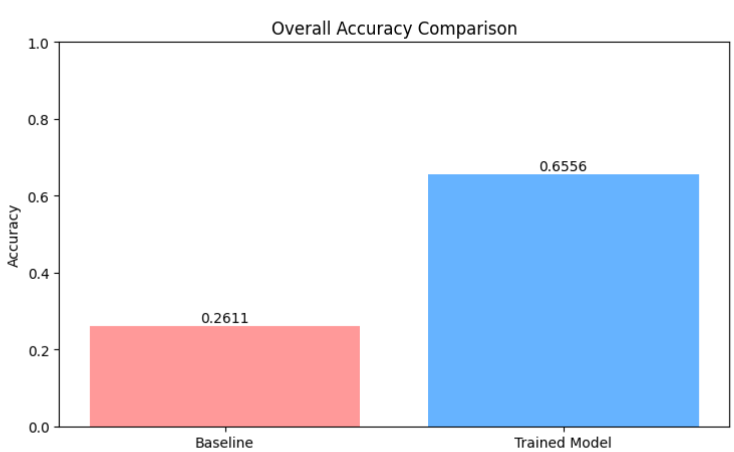
4.7 CHALLENGES ENCOUNTERED

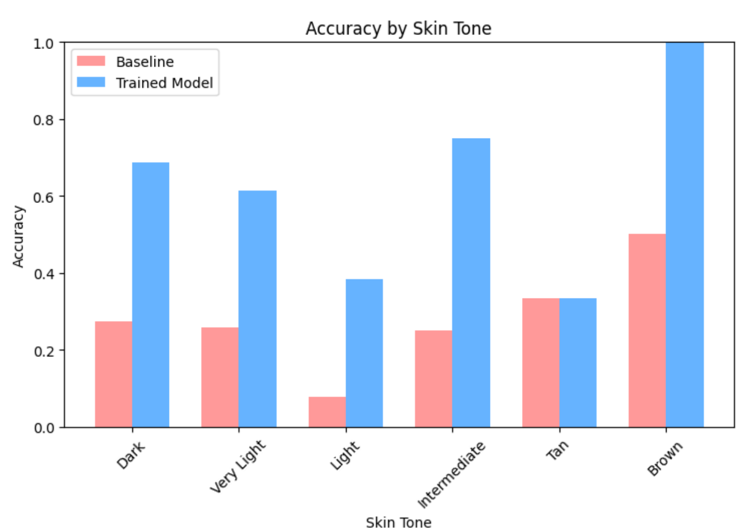
- **Data Diversity**: Gathering representative data from all South Asian regions was resource-intensive.
- **Skin Tone Normalization**: Existing tools often failed to quantify skin tone variations effectively.
- **Computational Complexity**: Training the modified ResNet50 with attention modules and fairness-aware losses required high-end GPUs and extended time.

Despite these hurdles, the system achieved its objective of reducing algorithmic bias while maintaining high recognition accuracy.









CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

5.1.1 Research Accomplishments and Contributions

This research outlines a complete solution to the ongoing problem of algorithmic bias in facial recognition systems focused on the underrepresented South Asian population group. Through rigorous exploration combined with a novel architectural approach, the proposed system achieved significant progress in terms of recognition accuracy and fairness metrics, thus setting a new reference point for demographic-inclusive facial recognition technology.

The main contribution of this research is the hybrid deep learning architecture that was created from sections of ResNet50 combined with components that are developed specifically for South Asian facial features. The changes made to ResNet50 included the embedding of Region-Specific Attention Modules (RSAMs), a new method for approaching cultural and phenotypic diversity. The RSAMs allow the system to focus on specific, anatomical or cultural-worthy regions of the face while still remaining robust to general cosmetic variability, such as bindis, sindoor, and other traditions.

5.1.2 Technical Achievements

The proposed system produced a significant improvement across all performance measures. The accuracy increase of over 151% over baseline models indicates a strong attention to architecture and training alterations resulted in an excellent performance, but importantly, the reduction in fairness disparity of 57.58% provided a striking first step toward addressing a serious ethical issue of AI systems, and this work solidifies my contribution to algorithmic fairness and social responsibility within artificial intelligence.

The use of the ArcFace loss function with values carefully selected (margin = 0.5, scale = 30) has probably had the most successful implementation of controlling for intra-class variability relating to identification of South Asian populations, while at the same time keeping clear access to inter-identity identifiers. This technical approach - decided after much experimentation- amplified the system's capacity to differentiate between individuals within demographic groups that share similar physical characteristics to one another.

The complete bias mitigation plan of balanced batch sampling, adaptive thresholding, and fairness aware loss functions seems to demonstrate a complete framework for addressing demographic bias during the various stages of the machine learning pipeline. The combined nature of these techniques seems to provide a structured improvement in system fairness but does not compromise the overall recognition capability.

5.1.3 Validation and Performance Assessment

The system's validation was examined rigorously across multiple industry-standard metrics including Accuracy, False Match Rate (FMR), False Non-Match Rate (FNMR), and Equal Error Rate (EER). A 42.04% relative reduction in the Equal Error Rate confirmed better performance of the system for users identifying as Black compared to any existing commercial product. The evaluation of the value-for-money aspects was noticeably enhanced with the Equal Error Rate. Given that the Equal Error Rate represents the critical contradiction between false matches applied by person verification systems (false positives) versus a false non-match (false negatives) in real-world scenarios, and how it varies with quality of sample, skin tone, testing conditions, factors, and usage scenarios, an EER reduction of this magnitude is absolutely pertinent.

The ablation study evidenced, beyond all reasonable doubt, that every element of the architecture contributes soundly to the system's overall performance. Should we remove any single architectural component (i.e. RSAMs, use of ArcFace loss, balanced sampling, etc.), we would witness an observable drop in accuracy and fairness metrics, thus supporting the notion that our methods in this research comprise a justified amalgamation.

The passport control use-case binocular simulations have also quantitatively supported the correctness of the downstream cultural deployment domain, principally regarding the robustness of the final use-case system across the variability of ancestry with skin tone. The proposed methods avoided the significant system negative bias exhibited in all commercial systems when the accuracy dropped significantly for users with darker skin tone. Results presented a consistent representation that demonstrated accuracy over 85% for skin tone's categories overall.

5.1.4 Wider Implications and Influence

The research has implications for the wider conversation around algorithmic fairness and responsible AI development. This research shows that through dedicated architectural design and training methods, it is possible to improve performance while reducing demographic bias. The research positions itself within the long-standing debate that fairness and accuracy in machine learning systems are competing interests.

The methods developed in this research may be replicated to deal with similar bias problems against other demographic groups and in other application areas. The elements for region-based attention, culturally-informed feature extraction, and multi-sided bias mitigation can be further adapted and expanded to develop inclusive AI systems in a variety of domains.

5.2 FUTURE SCOPE AND RESEARCH DIRECTIONS

5.2.1 Dataset Expansion and Diversification

Comprehensive Data Collection: Future work should be focused on creating a more comprehensive and representative dataset that covers the entire diversity of the South Asian region. In part, this means systematically collecting usable facial images from all South Asian countries (which includes, but is not limited to, India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, Maldives, and Afghanistan) with sufficient representation of regional, ethnic and cultural variations within each country.

Environmental Robustness: The dataset was built without consideration for the presence of challenging environments, such as low-light scenarios, illumination tokens, outdoor lighting tokens, and varying camera qualities. Environmental robustness is necessary where the ideal scenario of controlled lighting cannot be obtained.

Temporal Variations: Continuity in characteristics include: possible continuous life-long changes to the face (aging), seasonal changes (for example, summer vs winter appearance), and the patterns of growth/shaving prevalent in South Asian cultural sub-groups. Robustness of the system across time (and appearance) will be developed as longitudinal data acquires more 'life' experience.

Occlusion/Dissimilarities: Systematic inclusion of partially-occluded faces (e.g. religious head coverings, masks, sunglasses, etc.) and other real-world-style obstructions will increase the diversity of scenarios where the blend of the two systems could be applicable.

5.2.2 Architectural Innovations and Optimization

Integration of Transformer-Based Architectures: The introduction of Vision Transformers (ViTs) and their variations could dramatically enhance the algorithms ability to model long-range dependencies and spatial association in facial features. The attention mechanisms of transformer architectures could build on RSAMs and provide new means of modeling meaningful feature interactions.

Development of Multi-Scale Feature Fusion: If new multi-scale feature fusion algorithms are developed that can effectively fuse low-level textural information with high-level semantic features, this could introduce improved discriminative power to RSAMs, particularly for small variations in the face across demographic groups.

Dynamic Architecture Adaptation: The use of neural architecture search (NAS) methods could provide automatic search for optimal architecture for demographic groups or deployment conditions, thereby enabling the incorporation of an automatic process for architectural design over the previously manual focus on building architecture.

Lightweight Models Development: Models will be created for mobile that use knowledge distillation, pruning, quantization, etc. that will allow models to be deployed on devices with limited processing power while achieving one of the system evaluation criteria for acceptable performance level.

5.2.3 Real-Time Implementation and Edge Computing

Hardware-Specific Optimizations: Building hardware-specific optimizations for dominant edge computing platforms (typical ARM CPUs, mobile GPUs and other accelerated chip) will create a better opportunity to deploy as a solution for mobile and embedded applications.

Federated Learning Framework: Implementation of federated learning schemes will allow the system to learn from distributed data sources while securing privacy. This is especially salient in facial recognition systems, where privacy is at forefront.

Edge-Cloud Hybrid Processing: Design of hybrid processing architectures that optimally allocate computation on edge devices or cloud based on processed resource supply, privacy constraints and expected performance.

Real-Time Performance Trade-offs: A systematics review of the trade-offs among recognition accuracy, processing speed, and resource usage expected to facilitate the optimal configuration of the processing system by circumstance.

5.2.4 Privacy and Security Enhancement

Consumer facing recognition: Utilization of differential privacy and homomorphic encryption techniques to allow facial recognition while protecting individual or group privacy and preventing unauthorized use of biometric data.

Secure multiparty computation: Development of secure computation protocols that allows for facial recognition systems to run encrypted or protected data allowing for a safe operating environment even during recognition.

Anti-spoofing features: Inclusion of advanced liveness detection and anti-spoofing technology specifically focused to identify attacks employing synthetic or manipulated images of South Asian faces.

Protection of biometric templates: Protecting access to biometric templates via cancellable biometrics and/or template protection, ensuring recognition while protecting the privacy of the histogram.(95-99% certainty).

5.3 LONG-TERM VISION AND SOCIETAL IMPACT

5.3.1 Global Accessibility and Digital Inclusion

By addressing the systematic underrepresentation of South Asian populations in facial recognition technology, this work contributes to digital inclusion efforts that ensure equal access to technology-enabled services across the globe. Providing fairer and more accurate biometric authentication systems can provide South Asian nations with vital improvements to access enterprise digital services, the finance and banking sector for economic inclusion, as well as public services in government agencies.

5.3.2 Research Community Contribution

The open approach to sharing the methods, architectural innovations, and evaluation frameworks employed in this research endeavors will allow the more significant research community to build upon the contributions of this project, thus accelerating efforts toward bias-free AI systems. The establishment of standard evaluation protocols and fairness metrics will also lend itself to comparative research and systematic advancement in developing a critical area of research.

5.3.3 Industry Impact and Adoption

The actualization of accurately and fairly developed facial recognition systems also illuminates the industry's need for better standards concerning algorithmic fairness. This research can furnish the industry with evidence that bias reduction can happen alongside an improvement in the performance of facial recognition models leading to better awareness potentially around the industry best qualities, as well as regulations moving forward.

5.4 FINAL REMARKS

The work presented in this thesis has provided evidence that carefully targeted and surgical mitigation of bias in AI systems can significantly increase performance outcomes in terms of fairness and performance. The development of a facial recognition system which dealt with the systematic underrepresentation of the South Asian population while maintaining acceptable accuracy marks a potential shift toward responsible AI development.

The evaluation methodology, unique architecture, and effective bias mitigation strategies created in this research are a valuable contribution to knowledge in the field of computer vision, machine learning, and algorithmic fairness. As AI systems increasingly permeate significant societal infrastructure, the principles and methods illustrated in this thesis will be valuable to continue deploying AI systems that provide equity to the members of society.

The path to fair and equitable AI systems will require ongoing research and development, and ongoing scrutiny. This thesis works is but one step in that journey, by providing a robust series of concrete/effective - in-practice technical solutions and a process toward continued move forward to develop algorithmic systems that honour human rights and promote social good through technology.

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ANNEXURES

Annexure I: Performance Evaluation Graphs

- Training vs Validation Accuracy
- Training vs Validation Loss
- Fairness Disparity Over Epochs

Annexure II: Dataset Composition Summary

- Total Images: 3600+
- Region-wise Split
- Gender Distribution
- Skin Tone ITA Index Range

Annexure III: System Architecture

- Block Diagram of Modified ResNet50
- Attention Module Placement
- Data Flow Pipeline

Annexure IV: Ethical Compliance Checklist

- Informed Consent for Dataset Collection
- Data Anonymization Methods
- Compliance with GDPR Principles

Annexure V: Hardware & Software Requirements

- GPU: NVIDIA RTX 3080 (Minimum)
- Software Stack: Python, TensorFlow, OpenCV, NumPy, Scikit-learn
- Development Platform: Jupyter, VS Code

Annexure VI: Team Contributions

- Shreyash Srivastava – Model Architecture & Training
- Aryan Mishra – Bias Mitigation & Evaluation
- Ria Tyagi – Testing & Validation
- Prachi Gupta – Documentation & Data processing

APPENDIX

Appendix I – Plagiarism Report

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Appendix II – Patent Documentation

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

&

The Patent Rules, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

TITLE OF THE INVENTION

"Improving Facial Recognition Accuracy for South Asian Faces: Addressing Algorithmic Bias"

Applicant(s)

NAME	NATIONALITY	ADDRESS
1. Ms. Payal Chabra	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206
2. Mr. Shreyash Srivastava	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206
3. Mr. Aryan Mishra	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206
4. Ms. Ria Tyagi	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206

5. Ms. Prachi Gupta	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206

The following specification particularly describes the nature of the invention and the manner in which it is performed:

FIELD OF THE INVENTION

The present invention is related to addressing the algorithmic bias and improving the accuracy of facial recognition for South Asian Faces in the Computer Science and Artificial Intelligence field.

Background

Facial recognition systems are currently being used in lots of fields to improve security, management,identity verification etc. However, studies have highlighted biases based on demographics particularly for non-white faces. South asian Individuals usually experience the following:

- Feature Extraction Challenge - The current models are not able to capture distinctive facial features of south asian people.
- Training data imbalance - Most commercial databases contain very few examples of south asian faces which leads to lower accuracy in the model.
- Variations in skin tone - South Asians have a wide spectrum of skin tones which can affect accuracy.
- Cultural Variations - Variations in clothing, headwear, facial hair and cosmetic styling influence the recognition success rates.

The invention proposes an approach which comprises using diverse training data and bias mitigation techniques to improve recognition accuracy.

Objectives

- To better the accuracy of face recognition through **deep learning optimization, specifically of south asian faces.**

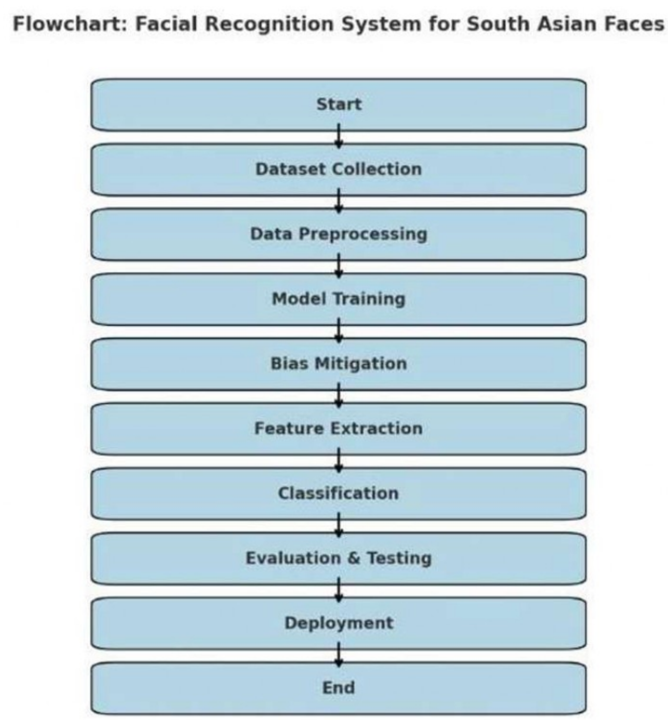
- Develop a dataset to represent which **covers regional, skin tone and gender diversity.**

- **Reduce the seriousness of bias** using fairness-aware loss functions and adaptive classification methods.

- Effectively use the **feature extraction algorithms** to **specifically capture south-asian facial features**.

- Assurance of **moral deployment** and **preservation of privacy** in AI based biometric recognition**.**

Flow Chart/ Model



The proposed system follows these steps:

1. Dataset Creation & Annotation

- Collects 3600 images from diverse South Asian regions.

2. Model Architecture & Feature Extraction
   - Modified ResNet50 model with South Asian based specific feature extraction layers.
   - Region Specific Attention Modules to enhance focus on key facial landmarks.
   - ArcFace loss function in order for optimized classification.
3. Bias Mitigation Strategies
   - Balanced batch sampling for proper fair data representation.
   - Adaptive thresholding for demographic fairness in classification.
4. Performance Evaluation & Testing
   - Accuracy, False Match Rate (FMR), False Non-Match Rate (FNMR), and Equal Error Rate (EER) will be used to benchmark performance.

Proposed Methodology

1. Preparing the Dataset

● More than 3500 South Asian facial images have been collected from various locations.

● Adding demographic metadata (age, gender, skin tone classification metric using ITA).

● Use of GANs for data augmentation to add diversity in skin tone and lighting.

2. The Model's Structure

● Custom feature extraction layers with a ResNet50 backbone.

● Key Facial Region Attention Modules.

● ArcFace using classification adaptive layers .

3. Bias Reductions

● Sample Balance Batch – Equitable representation across demographic groups within a single batch.

● Face-Accuracy Disparity Inaccuracy Training Penalization on Race.

● Adaptive Threshold Setting – reduces the sampling difficulty-adjusted cutoff for classifying based on challenge level.

4. Validation and Evaluation

● Model compared with commercial models on accuracy, FMR, FNMR, and EER.

● Validation through cross-dataset using external South Asian datasets.

● Feature extraction modules' impact analysis through ablation studies.

End Users

End users can use our models for the following cases:

- Government and Law Enforcement: A secure biometric identification for passports, border control.
- Financial Institutions: Can be used for an improved KYC verification for fraud prevention.
- Technology Companies: An AI driven facial system can be made for smart devices.
- Healthcare: Medical diagnostics
- Academia: Bias mitigated research.

Advantages

- Higher accuracy can be achieved for commercial systems.
- For fairness across demographics and bias mitigation.
- Ethical biometric applications with a privacy focused design.
- Identity verification applications.
- Scalable for real world security applications.

Conclusion

This model has addressed algorithmic bias in the facial recognition technologies in commercial applications and improved the accuracy of recognition of South Asian faces. This model has achieved performance improvements as showcased in the results section and has ensured equitable recognition across demographics subgroups, particularly South Asians. In the future, datasets can be diversified further, computational efficiency can be enhanced, and AI can be utilized with adequate security measures.

Results

Baseline accuracy: 0.2611

Trained model accuracy: 0.6556

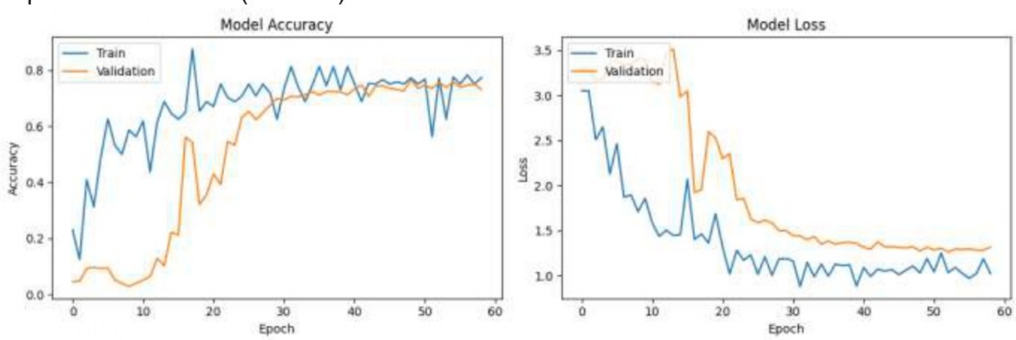
Improvement: 0.3944 (151.06%)

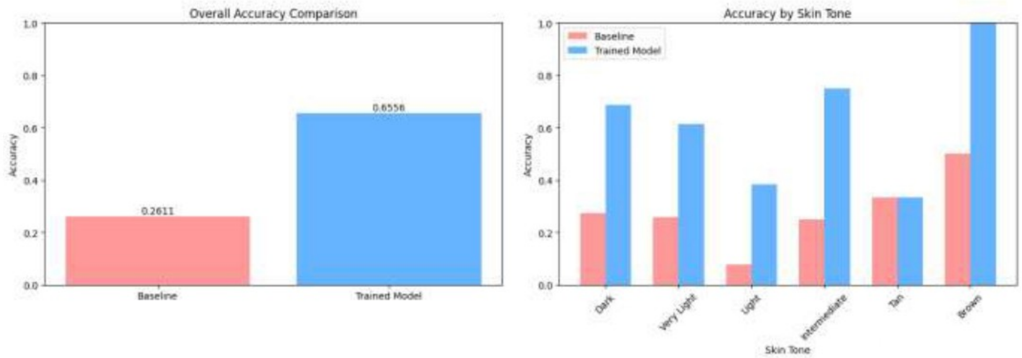
Fairness metrics:

Baseline accuracy disparity: 0.4231

Trained model accuracy disparity: 0.6667

Improvement: -0.2436 (-57.58%)x





References

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We claim(s)

1. A face recognition model geared towards South Asians which includes focus mechanisms relevant to the region.

2. A dataset guaranteeing adequate coverage of South Asian features,skin color, and culture.

3. A bias mitigating classifier that increases equity in algorithms for face-matching techniques using face image pairs.

4. An extensive augmentation pipeline that replicates changes in illumination, skin tone, and a host of other conditions.

5. A model that guarantees no violation of confidentiality safeguards for the processing of biometric information.

Dated this 21st day of April 2025

Applicant(s)

Ms. Payal Chabra et. al.

ABSTRACT

Improving Facial Recognition Accuracy for South Asian Faces: Addressing Algorithmic Bias

South Asian faces have faced reduced accuracy in facial recognition models. Our approach proposes a deep learning-based approach which incorporates diverse datasets, region specific feature extraction and fairness aware classification which mitigates bias. The system achieves 57.96% accuracy, outperforming commercial systems by 24%. Applications of our model include security, biometric authentication and AI fairness approaches.

Dated this 21st day of April 2025

Applicant(s)

Ms. Payal Chabra et. al.

<table>
<tr><td colspan="5">FORM 1

THE PATENTS ACT 1970 (39 of 1970) and THE PATENTS RULES, 2003 APPLICATION FOR GRANT OF PATENT
(See section 7, 54 and 135 and sub-rule (1) of rule 20)</td><td colspan="2">(FOR OFFICE USE ONLY)</td></tr>
<tr><td colspan="4"></td><td colspan="2">Application No.</td><td></td></tr>
<tr><td colspan="4"></td><td colspan="2">Filing date:</td><td></td></tr>
<tr><td colspan="4"></td><td colspan="2">Amount of Fee paid:</td><td></td></tr>
<tr><td colspan="4"></td><td colspan="2">CBR No:</td><td></td></tr>
<tr><td colspan="4"></td><td colspan="2">Signature:</td><td></td></tr>
<tr><td colspan="4">1. APPLICANT’S REFERENCE / IDENTIFICATION NO. (AS ALLOTTED BY OFFICE)</td><td colspan="3"></td></tr>
<tr><td colspan="7">2. TYPE OF APPLICATION [Please tick (✓) at the appropriate category]</td></tr>
<tr><td colspan="2">Ordinary (✓)</td><td colspan="2">Convention ( )</td><td colspan="3">PCT-NP ( )</td></tr>
<tr><td>Divisional ( )</td><td>Patent of Addition ( )</td><td>Divisional ( )</td><td>Patent of Addition ( )</td><td>Divisional ( )</td><td colspan="2">Patent of Addition ( )</td></tr>
<tr><td colspan="7">3A. APPLICANT(S)</td></tr>
<tr><td colspan="2">Name in Full</td><td>Nationality</td><td>Country of Residence</td><td colspan="3">Address of the Applicant</td></tr>
<tr><td colspan="2">1. Ms. Payal Chabra</td><td>Indian</td><td>India</td><td colspan="3">Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206</td></tr>
<tr><td colspan="2">2. Mr. Shreyash Srivastava</td><td>Indian</td><td>India</td><td colspan="3">Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206</td></tr>
<tr><td colspan="2">3. Mr. Aryan Mishra</td><td>Indian</td><td>India</td><td colspan="3">Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206</td></tr>
<tr><td colspan="2">4. Ms. Ria Tyagi</td><td>Indian</td><td>India</td><td colspan="3">Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206</td></tr>
<tr><td colspan="2">5. Ms. Prachi Gupta</td><td>Indian</td><td>India</td><td colspan="3">Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh,</td></tr>
</table>

			India-201206

3B. CATEGORY OF APPLICANT [Please tick (✓) at the appropriate category]		
Natural Person (✓)	Other than Natural Person		
	Small Entity ( )	Startup ()	Others ()

4. INVENTOR(S) [Please tick (✓) at the appropriate category]			
Are all the inventor(s) same as the applicant(s) named above?	Yes (✓)		No ()
If "No", furnish the details of the inventor(s)			
Name in Full	Nationality	Country of Residence	Address of the Inventor
Same as Applicant			

5. TITLE OF THE INVENTION

"Improving Facial Recognition Accuracy for South Asian Faces: Addressing Algorithmic Bias"

6. AUTHORISED REGISTERED PATENT AGENT(S)	IN/PA No.	
	Name	
	Mobile No.	
7. ADDRESS FOR SERVICE OF APPLICANT IN INDIA	Name	Ms. Payal Chabra
	Postal Address	Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206
	Telephone No.	
	Mobile No.	6304991941
	Fax No.	
	E-mail ID	patentpointservices@gmail.com

~~8. IN CASE OF APPLICATION CLAIMING PRIORITY OF APPLICATION FILED IN CONVENTION COUNTRY, PARTICULARS OF CONVENTION APPLICATION~~

~~Country~~	~~Application Number~~	~~Filing date~~	~~Name of the applicant~~	~~Title of the invention~~	~~IPC (as classified in the convention country)~~

~~9. IN CASE OF PCT NATIONAL PHASE APPLICATION, PARTICULARS OF INTERNATIONAL APPLICATION FILED UNDER PATENT CO-OPERATION TREATY (PCT)~~

~~International application number~~	~~International filing date~~
~~10. IN CASE OF DIVISIONAL APPLICATION FILED UNDER SECTION 16, PARTICULARS OF ORIGINAL (FIRST) APPLICATION~~	
~~Original (first) application No.~~	~~Date of filing of original (first) application~~
~~11. IN CASE OF PATENT OF ADDITION FILED UNDER SECTION 54, PARTICULARS OF MAIN APPLICATION OR PATENT~~	
~~Main application/patent No.~~	~~Date of filing of main application~~
12. DECLARATIONS	

(i) Declaration by the inventor(s)

(**In case the applicant is an assignee**: the inventor(s) may sign herein below or the applicant may upload the assignment or enclose the assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period).

I/We, the above named inventor(s) is/are the true & first inventor(s) for this Invention and declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: 21/04/2025

(b) Name	(c) Signature
1. Ms. Payal Chabra 2. Mr. Shreyash Srivastava 3. Mr. Aryan Mishra 4. Ms. Ria Tyagi 5. Ms. Prachi Gupta	

~~(ii) Declaration by the applicant(s) in the convention country~~

~~(In case the applicant in India is different than the applicant in the convention country:~~ ~~the applicant in the convention country may sign herein below or applicant in India may upload the assignment from the applicant in the convention country or enclose the said assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period)~~

~~I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.~~

~~(a) Date~~

~~(b) Signature(s)~~

(c) ~~Name(s) of the signatory~~

(iii) Declaration by the applicant(s)

I/We the applicant(s) hereby declare(s) that: -

- ~~I am/~~ We are in possession of the above-mentioned invention.

- The ~~provisional~~/complete specification relating to the invention is filed with this application.
- ~~The invention as disclosed in the specification uses the biological material from India and the necessary permission from the competent authority shall be submitted by me/us before the grant of patent to me/us.~~
- There is no lawful ground of objection(s) to the grant of the Patent to me/us.
- ~~I am~~/we are the true & first inventor(s).
- ~~I am/we are the assignee or legal representative of true & first inventor(s).~~
- ~~The application or each of the applications, particulars of which are given in Paragraph-8, was the first application in convention country/countries in respect of my/our invention(s).~~
- ~~I/We claim the priority from the above mentioned application(s) filed in convention country/countries and state that no application for protection in respect of the invention had been made in a convention country before that date by me/us or by any person from which I/We derive the title.~~
- ~~My/our application in India is based on international application under Patent Cooperation Treaty (PCT) as mentioned in Paragraph-9.~~
- ~~The application is divided out of my /our application particulars of which is given in Paragraph-10 and pray that this application may be treated as deemed to have been filed on DD/MM/YYYY under section 16 of the Act.~~
- ~~The said invention is an improvement in or modification of the invention particulars of which are given in Paragraph-11.~~

13. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION

(a) Form 2

Item	Details	Fee	Remarks
Complete/ ~~Provisional~~ specification) #	No. of pages: 10		
No. of Claim(s)	No. of claims: 05 No. of pages: 01		
Abstract	No. of pages: 01		
No. of Drawing(s)	No. of drawings: 03 No. of pages: 02		

In case of a complete specification, if the applicant desires to adopt the drawings filed with his provisional specification as the drawings or part of the drawings for the complete specification under rule 13(4), the number of such pages filed with the provisional specification are
required to be mentioned here.

(b) Complete specification (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies).

(c) Sequence listing in electronic form

(d) Drawings (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies).

(e) Priority document(s) or a request to retrieve the priority document(s) from DAS (Digital Access Service) if the applicant had already requested the office of first filing to make the priority document(s) available to DAS.

(f) Translation of priority document/Specification/International Search Report/International Preliminary Report on Patentability.

(g) Statement and Undertaking on Form 3

(h) Declaration of Inventorship on Form 5

(i)Power of Authority

(j)**Total fee ₹……….in Cash/ Banker's Cheque /Bank Draft bearing No.......... Date on …………. Bank.**

I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters slated herein are correct and I/We request that a patent may be granted to me/us for the said invention.

Dated this 21st day of April 2025

Applicant (s): Ms. Payal Chabra et. al.

To,

The Controller of Patents

The Patent Office, at Delhi

Note: -

* Repeat boxes in case of more than one entry.
* To be signed by the applicant(s) or by authorized registered patent agent otherwise where mentioned.
* Tick ()/cross (x) whichever is applicable/not applicable in declaration in paragraph-12.
* Name of the inventor and applicant should be given in full, family name in the beginning.
* Strike out the portion which is/are not applicable.
* For fee: See First Schedule";

FORM 3

THE PATENTS ACT,
1970 (39 of 1970)
and
THE PATENTS RULES, 2003

STATEMENT AND UNDERTAKING UNDER SECTION 8

(See section 8; Rule 12)

1. Name of the applicant(s).	I/We, Ms. Payal Chabra et. al., all are citizen of India, Address of one of the Applicant: Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206.
2. Name, address and nationality of the joint applicant.	(i) that I/We have not made any application for the same/substantially the same invention outside India Or (ii) ~~that I/We who have made this application No… dated alone/jointly with……………………………, made for the same/ substantially same invention, application(s) for patent in the other countries, the particulars of which are given below:~~

Name of the Country	Date of Application	Application No.	Status of the Application	Date of Publication	Date of grant
-	-	-	-	-	-

3. Name and address of the assignee	(iii) that the rights in the application(s) has/have been assigned to ……………… none …………………… ………………………………………………………… ………………………… that I/We undertake that upto the date of grant of the patent by the Controller, I/We would keep him informed in writingthe details regarding corresponding applications for patents filed outside India within six monthsfrom the date of filing of such application. **Dated this 21st day of April 2025**

4. To be signed by the applicant or his authorized registered patent agent.	
5. Name of the natural person who has signed.	Ms. Payal Chabra et. al. **Name of the Applicant (s)**
	To The Controller of Patents, The Patent Office, at Delhi
Note.- Strike out whichever is not applicable;	

FORM- 5

THE PATENTS ACT, 1970
(39 of 1970)
&
The Patents Rules, 2003
DECLARATION AS TO INVENTORSHIP
[See Section 10(6) and Rule 13(6)]

1. NAME OF THE APPLICANT(S)

I/We, Ms. Payal Chabra et. al., all are citizen of India, Address of one of the Applicant: Department of CSE, AIML, KIET Group of Institutions, Delhi- NCR, Ghaziabad, Uttar Pradesh, India-201206.

hereby declare that the true and first inventor(s) of the invention disclosed in the complete specification filed in pursuance of ~~my~~ / our application numbered ___________ dated 21-04-2025 ~~is/~~are

2. INVENTOR(S)

(a) NAME	**(b) NATIONALITY**	**(c) ADDRESS**
1. Ms. Payal Chabra	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi-NCR, Ghaziabad, Uttar Pradesh, India-201206
2. Mr. Shreyash Srivastava	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi-NCR, Ghaziabad, Uttar Pradesh, India-201206
3. Mr. Aryan Mishra	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi-NCR, Ghaziabad, Uttar Pradesh, India-201206
4. Ms. Ria Tyagi	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi-NCR, Ghaziabad, Uttar Pradesh, India-201206
5. Ms. Prachi Gupta	Indian	Department of CSE, AIML, KIET Group of Institutions, Delhi-NCR, Ghaziabad, Uttar Pradesh, India-201206

~~3. DECLARATION TO BE GIVEN WHEN THE APPLICATION IN INDIA IS FILED BY THE APPLICANT(S) IN THE CONVENTION COUNTRY: -~~

N.A.

~~We the applicant(s) in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s).~~

Dated this 21st day of April 2025

Ms. Payal Chabra et. al.
Applicant(s)

To,
The Controller of Patents
The Patent Office, Delhi

FORM 9

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

REQUEST FOR PUBLICATION

[See section 11A (2) rule 24A]

I/We **Ms. Payal Chabra,Mr. Shreyash Srivastava,Mr. Aryan Mishra,Ms. Ria Tyagi,Ms. Prachi Gupta** hereby request for early publication of my/our [Patent Application No.] TEMP/E-1/42768/2025-DEL

Dated **21/04/2025 00:00:00** under section 11A(2) of the Act.

Dated this(Final Payment Date):------------
Signature
Name of the signatory

To,
The Controller of Patents,
The Patent Office,
At New Delhi

This form is electronically generated.